Lesson 07 Demo 01

JSON Response from the API

Objective: To create a React application using a component that makes an API call and displays the retrieved data

Tools Required: Visual Studio Code

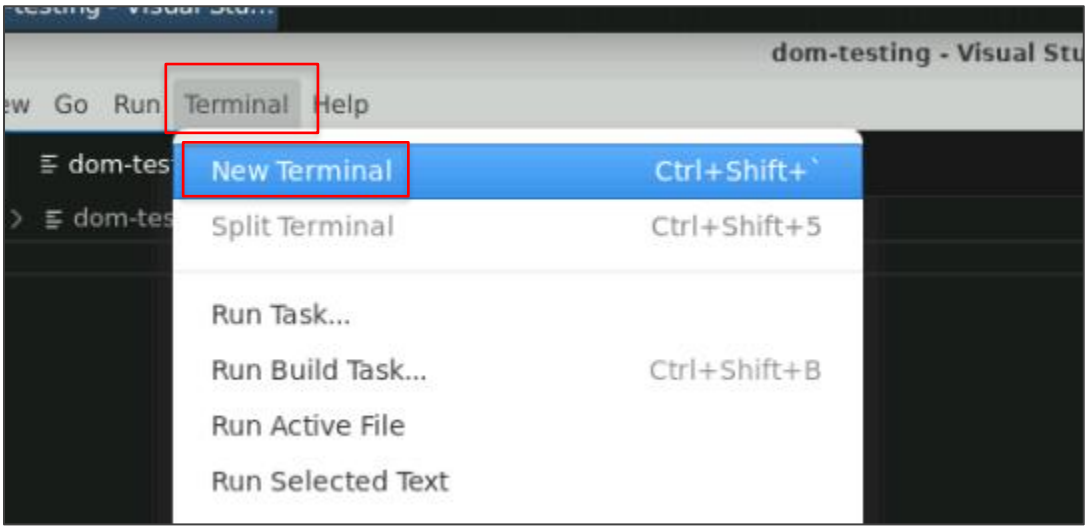
Prerequisites: NA

Steps to be followed:

1. Set up a new React project using Vite
2. Import and use the component
3. Run and verify the application

Step 1: Set up a new React project using Vite

1.1 Open Visual Studio Code, click on **Terminal,** and then select **New Terminal**



1.2 Run the following command to create a new React project using Vite:

npm create vite@latest api-demo

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

labuser@ip-172-31-12-132:~$ npm create vite@latest api-demo
```

1.3 Select **React** and press enter

```
> npx
> create-vite api-demo

◆ Select a framework:
  ○ Vanilla
  ○ Vue
  ● React
  ○ Preact
  ○ Lit
  ○ Svelte
  ○ Solid
  ○ Qwik
  ○ Angular
  ○ Others
```

1.4 Select **JavaScript** and press enter

```
> npx
> create-vite api-demo

◇  Select a framework:
   React

◆  Select a variant:
   ○ TypeScript
   ○ TypeScript + SWC
   ● JavaScript
   ○ JavaScript + SWC
   ○ React Router v7 ↗
```

1.5 Run the following command to navigate to the project folder:
cp api-demo/

```
● labuser@ip-172-31-10-95:~$ cd api-demo/
○ labuser@ip-172-31-10-95:~/api-demo$
```

1.6 Run the following command to install dependencies:
npm i

```
● labuser@ip-172-31-12-132:~/api-demo$ npm i

added 149 packages, and audited 150 packages in 21s

30 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

1.7 Run the following command to start the development server:

npm run dev

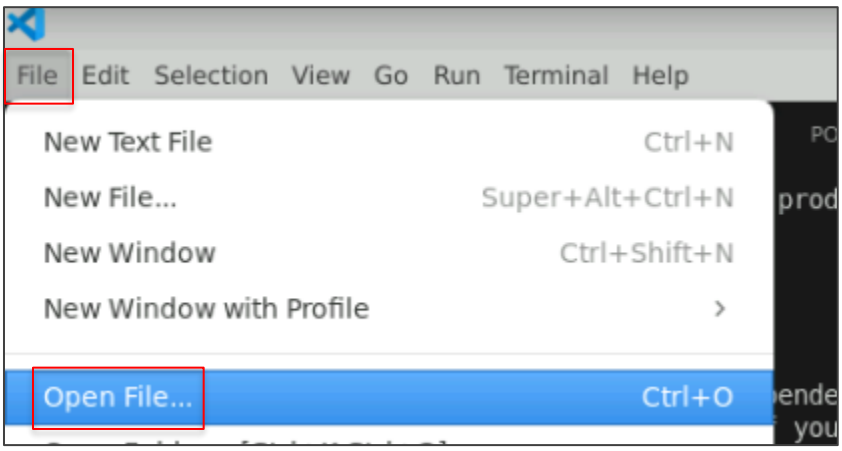
```
labuser@ip-172-31-12-132:~/api-demo$ npm run dev

> api-demo@0.0.0 dev
> vite


  VITE v6.2.2  ready in 408 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
```

1.8 Navigate to the web browser and paste the link given below:

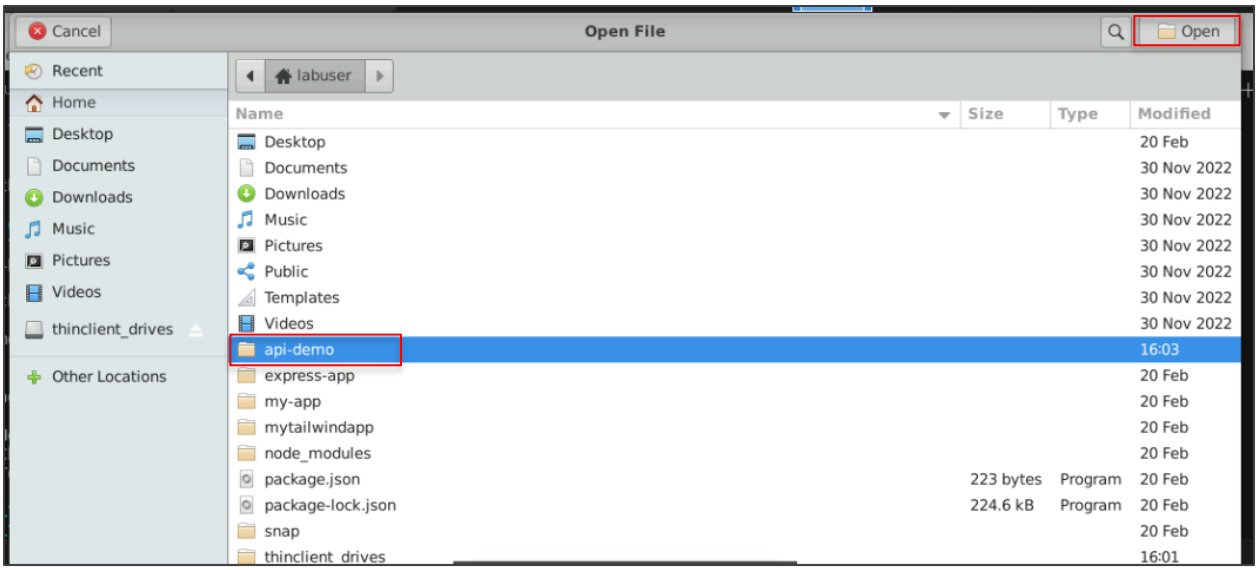
http://localhost:5173

Step 2: Import and use the component

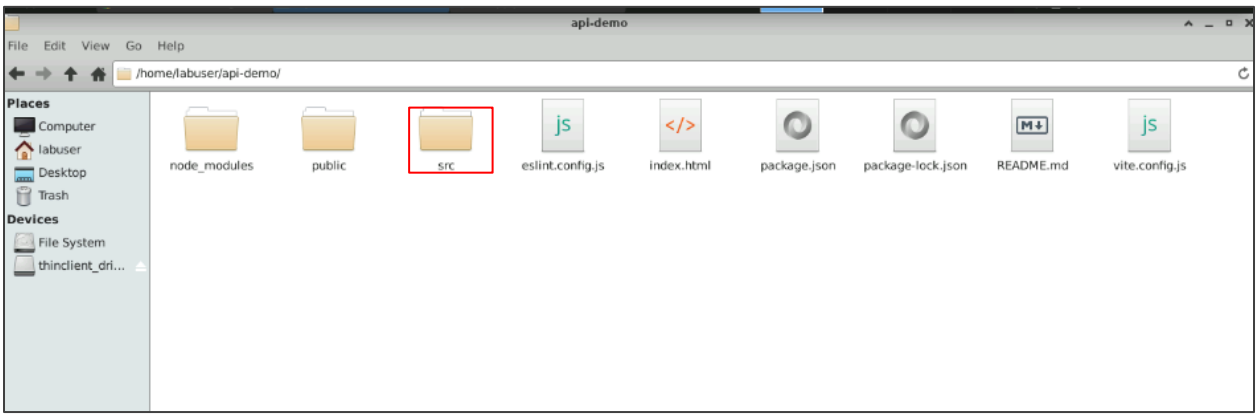
2.1 Open Visual Studio Code, click on **File**, and select **Open File…**



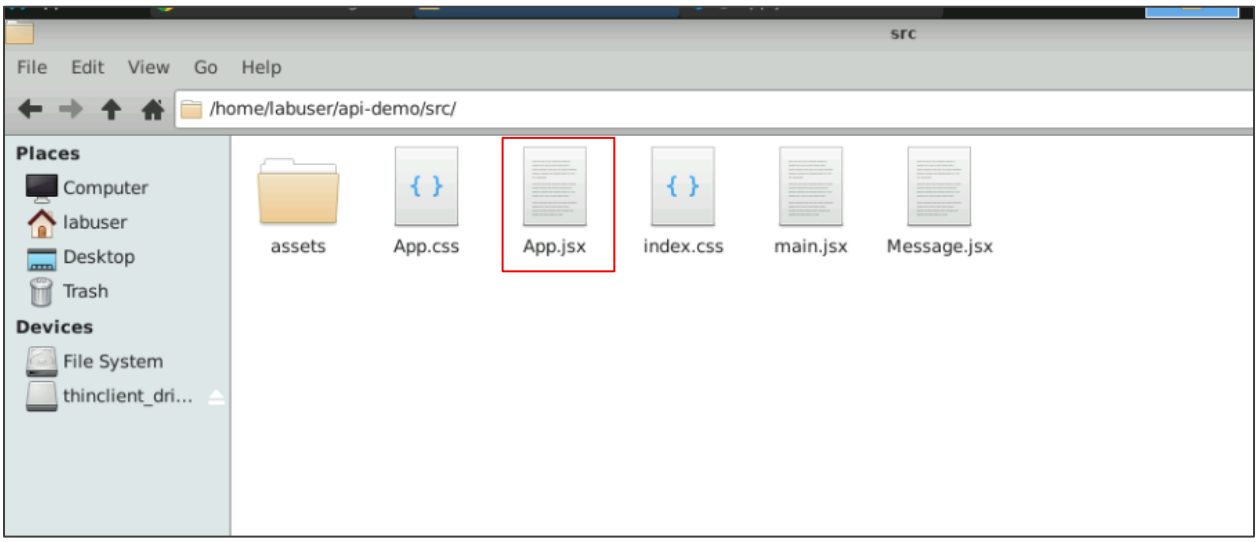
2.2 Select **api-demo** and click **Open**



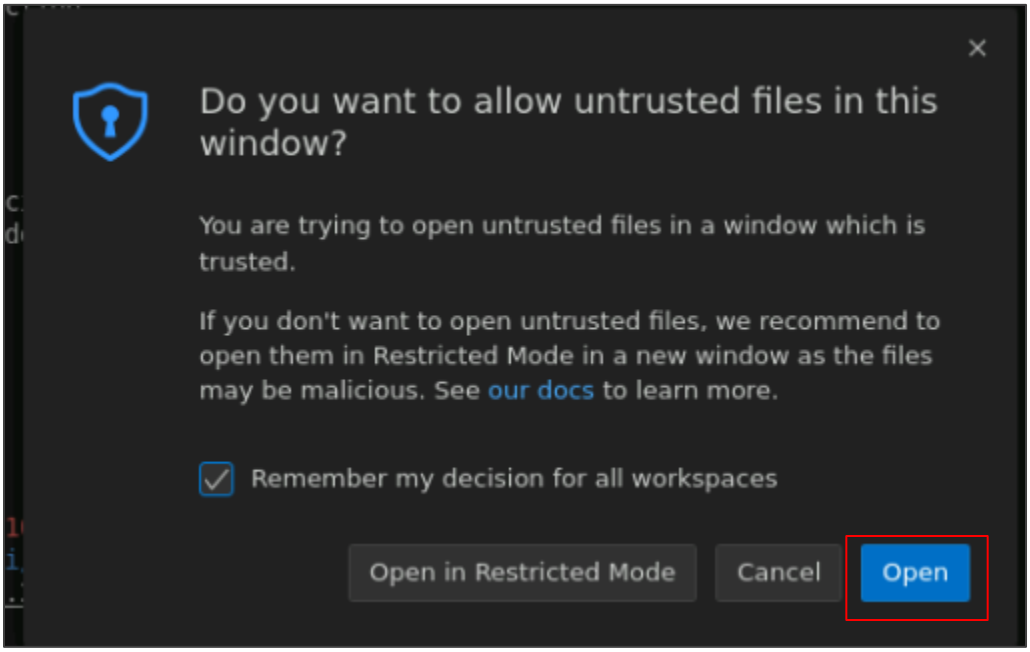
2.3 Double-click **src**



2.4 Open **App.jsx**



2.5 Click **Open**



2.6 Paste the code given below to include the new component in **App.jsx:**

```
import React, { useState, useEffect } from 'react';
import './App.css';

function App() {
  const [data, setData] = useState(null);

  async function fetchData() {
    const response = await fetch('https://jsonplaceholder.typicode.com/todos/1');
    const jsonData = await response.json();
    setData(jsonData);
  }

  useEffect(() => {
    fetchData();
  }, []);

  return (
    <div className="App">
      <h1>API Demo</h1>
```

```
    {data ? (
      <p>API response: {JSON.stringify(data)}</p>
    ) : (
      <p>Loading...</p>
    )}
  </div>
 );
}

export default App;
```

home > labuser > api-demo > src > App.jsx > ...

```
1   import React, { useState, useEffect } from 'react';
2   import './App.css';
3
4   function App() {
5     const [data, setData] = useState(null);
6
7     async function fetchData() {
8       const response = await fetch('https://jsonplaceholder.typicode.com/todos/1
9       const jsonData = await response.json();
10      setData(jsonData);
11    }
12
13    useEffect(() => {
14      fetchData();
15    }, []);
16
17    return (
18      <div className="App">
19        <h1>API Demo</h1>
```

Step 3: Run and verify the application

3.1 Navigate to the browser and refresh the web page

By following the step-by-step process outlined in this demo, you have successfully created a React application using Vite. You set up the project, built a React component that made an API call, and displayed the retrieved data dynamically. This hands-on experience provided insight into essential React concepts such as component structure, state management, and effect hooks for handling asynchronous operations.